

Problem Statement

Scaler is an online tech-versity offering intensive computer science & Data Science courses through live classes delivered by tech leaders and subject matter experts. The meticulously structured program enhances the skills of software professionals by offering a modern curriculum with exposure to the latest technologies. It is a product by InterviewBit.

You are working as a data scientist with the analytics vertical of Scaler, focused on profiling the best companies and job positions to work for from the Scaler database. You are provided with the information for a segment of learners and tasked to cluster them on the basis of their job profile, company, and other features. Ideally, these clusters should have similar characteristics.

Downloading the Dataset

```
!wget "https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/002/856/original/scaler_clustering.csv" -O scaler.csv

--2024-11-29 10:54:06-- https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/002/856/original/scaler_clustering.csv
Resolving d2beiqkhq929f0.cloudfront.net (d2beiqkhq929f0.cloudfront.net)... 18.64.229.71, 18.64.229.91, 18.64.229.172, ..
Connecting to d2beiqkhq929f0.cloudfront.net (d2beiqkhq929f0.cloudfront.net)|18.64.229.71|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 24735965 (24M) [text/plain]
Saving to: 'scaler.csv'

scaler.csv          100%[=====>] 23.59M  132MB/s   in 0.2s

2024-11-29 10:54:06 (132 MB/s) - 'scaler.csv' saved [24735965/24735965]
```

Basic checks on the data

```
import numpy as np
import pandas as pd
```

```
df= pd.read_csv("scaler.csv")
df.head()
```

|   | Unnamed: 0 | company_hash                    | email_hash                                        | orgyear | ctc     | job_position       | ctc_updated_year |
|---|------------|---------------------------------|---------------------------------------------------|---------|---------|--------------------|------------------|
| 0 | 0          | atrgxnnt xzaxv                  | 6de0a4417d18ab14334c3f43397fc13b30c35149d70c05... | 2016.0  | 1100000 | Other              | 2020.0           |
| 1 | 1          | qtrxvzwt<br>xzegwgbb<br>rxbxnta | b0aaf1ac138b53cb6e039ba2c3d6604a250d02d5145c10... | 2018.0  | 449999  | FullStack Engineer | 2019.0           |
| 2 | 2          | ojzwnvwnxw vx                   | 4860c670bcd48fb96c02a4b0ae3608ae6fdd98176112e9... | 2015.0  | 2000000 | Backend Engineer   | 2020.0           |

```
df.shape
```

```
(205843, 7)
```

Data Dictionary:

- 'Unnamed 0' - Index of the dataset
- Email\_hash - Anonymised Personal Identifiable Information (PII)
- Company\_hash - This represents an anonymized identifier for the company, which is the current employer of the learner.
- orgyear - Employment start date
- CTC - Current CTC
- Job\_position - Job profile in the company
- CTC\_updated\_year - Year in which CTC got updated (Yearly increments, Promotions)

```
df.isna().sum()
```

```
0
Unnamed: 0      0
company_hash    44
email_hash      0
orgyear        86
ctc             0
job_position   52564
ctc_updated_year 0
dtype: int64
```

```
df.nunique()
```

```
0
Unnamed: 0    205843
company_hash   37299
email_hash    153443
orgyear        77
ctc           3360
job_position   1016
ctc_updated_year 7
dtype: int64
```

Regex to clean the strings in the data.

```
import re
def remove_special(str):
    new_str=re.sub('[^A-Za-z0-9]+',' ',str).lower().strip()
    return new_str
```

Drop unnecessary columns

```
# Drop email_hash and unnamed features
df.drop(["Unnamed: 0","email_hash"],axis=1,inplace=True)
df.head()
```

```
0      company_hash  orgyear    ctc  job_position  ctc_updated_year
0      atrgxntt xzaxv   2016.0  1100000      Other          2020.0
1  qtrxvzwt xzegwgb rxbxnta  2018.0   449999  FullStack Engineer          2019.0
2      ojzwnvwnxw vx   2015.0  2000000  Backend Engineer          2020.0
3      ngpgutaxv   2017.0   700000  Backend Engineer          2019.0
4      qxen sqghu   2017.0  1400000  FullStack Engineer          2019.0
```

```
df
```



|        | company_hash             | orgyear | ctc     | job_position       | ctc_updated_year |   |
|--------|--------------------------|---------|---------|--------------------|------------------|-------------------------------------------------------------------------------------|
| 0      | atrgxnnt xzaxv           | 2016.0  | 1100000 | Other              | 2020.0           |  |
| 1      | qtrxvzwt xzegwgb rxbxnta | 2018.0  | 449999  | FullStack Engineer | 2019.0           |  |
| 2      | ojzwnvwnxw vx            | 2015.0  | 2000000 | Backend Engineer   | 2020.0           |                                                                                     |
| 3      | ngpgutaxv                | 2017.0  | 700000  | Backend Engineer   | 2019.0           |                                                                                     |
| 4      | qxen sqghu               | 2017.0  | 1400000 | FullStack Engineer | 2019.0           |                                                                                     |
| ...    | ...                      | ...     | ...     | ...                | ...              |                                                                                     |
| 205838 | vuurt xzw                | 2008.0  | 220000  | NaN                | 2019.0           |                                                                                     |
| 205839 | husqvawgb                | 2017.0  | 500000  | NaN                | 2020.0           |                                                                                     |
| 205840 | vwwgrxnt                 | 2021.0  | 700000  | NaN                | 2021.0           |                                                                                     |
| 205841 | zgn vuurxwvmrt           | 2019.0  | 5100000 | NaN                | 2019.0           |                                                                                     |
| 205842 | bgqsvz onvzrtj           | 2014.0  | 1240000 | NaN                | 2016.0           |                                                                                     |

205843 rows x 5 columns

Check for nulls in the dataset.

```
df['job_position'].isna().sum()
```

 52564

```
# Clean strings in company_hash and job_position
df["company_hash"]=df["company_hash"].apply(lambda x: remove_special(str(x)))
df["job_position"]=df["job_position"].apply(lambda x: remove_special(str(x)))
```

Cleaning and processing job\_position feature

```
df['job_position'].value_counts()[30:60]
```



|                                   | count |
|-----------------------------------|-------|
| job_position                      |       |
| intern                            | 42    |
| associatesoftwareengineer         | 35    |
| engineer                          | 29    |
| student                           | 25    |
| softwaredevelopmentengineer       | 24    |
| consultant                        | 24    |
| researchengineer                  | 22    |
| memberoftechnicalstaff            | 21    |
| projectengineer                   | 19    |
| softwaredevelopmentengineeri      | 16    |
| softwareengineerii                | 16    |
| assistantsystemengineer           | 15    |
| programmeranalyst                 | 15    |
| seniorengineer                    | 15    |
| softwaredevelopmentengineerii     | 14    |
| seniorsoftwaredeveloper           | 14    |
| leadsoftwareengineer              | 14    |
| softwaredeveloper                 | 13    |
| associate                         | 13    |
| dataengineer                      | 13    |
| associateconsultant               | 12    |
| technologyanalyst                 | 12    |
| seniorsystemsengineer             | 12    |
| seniorconsultant                  | 12    |
| softwareengineerfullstack         | 11    |
| teamlead                          | 11    |
| seniorsoftwaredevelopmentengineer | 10    |
| mts                               | 10    |
| softwareengineeri                 | 10    |
| seniorsoftwareengineerbackend     | 10    |

dtype: int64

```
df.duplicated().sum()
```



17624

```
df.drop_duplicates(inplace=True)
df.shape
```



(188219, 5)

```
# Removing rows that do not have company or job position details
df = df[~((df["company_hash"]=="")|(df["job_position"]==""))]
df.shape
```



(188126, 5)

```
df.isna().sum()
```

```
↵
```

|                  | 0  |
|------------------|----|
| company_hash     | 0  |
| orgyear          | 86 |
| ctc              | 0  |
| job_position     | 0  |
| ctc_updated_year | 0  |

dtype: int64

```
# Filling null values in orgyear with the median values based on company_hash
```

```
df['orgyear']=df['orgyear'].fillna(df.groupby("company_hash")["orgyear"].transform('median'))
df['orgyear'].isna().sum()
```

```
↵ <ipython-input-100-4d1157e3479b>:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user\_guide/indexing.html#returning-a-view-versus-a-copy
df['orgyear']=df['orgyear'].fillna(df.groupby("company_hash")["orgyear"].transform('median'))
26
```

```
# Removing rows where orgyear is null even after filling with median
```

```
df= df[~(df['orgyear'].isna())]
df.isna().sum()
```

```
↵
```

|                  | 0 |
|------------------|---|
| company_hash     | 0 |
| orgyear          | 0 |
| ctc              | 0 |
| job_position     | 0 |
| ctc_updated_year | 0 |

dtype: int64

```
df['orgyear'].value_counts()
```

**count****orgyear**

|        |       |
|--------|-------|
| 2018.0 | 22151 |
| 2016.0 | 21476 |
| 2017.0 | 21320 |
| 2015.0 | 19608 |
| 2019.0 | 18545 |
| 2014.0 | 16091 |
| 2013.0 | 12053 |
| 2020.0 | 10429 |
| 2012.0 | 10257 |
| 2011.0 | 7802  |
| 2010.0 | 5646  |
| 2009.0 | 3738  |
| 2021.0 | 3141  |
| 2008.0 | 2688  |
| 2007.0 | 2230  |
| 2006.0 | 2066  |
| 2005.0 | 1859  |
| 2004.0 | 1448  |
| 2003.0 | 1010  |
| 2022.0 | 836   |
| 2001.0 | 708   |
| 2002.0 | 683   |
| 2000.0 | 491   |
| 1999.0 | 338   |
| 1998.0 | 279   |
| 2023.0 | 245   |
| 1997.0 | 234   |
| 1996.0 | 134   |
| 1995.0 | 94    |
| 1991.0 | 79    |
| 1993.0 | 72    |
| 1994.0 | 65    |
| 1992.0 | 47    |
| 2024.0 | 43    |
| 1990.0 | 38    |
| 1989.0 | 22    |
| 0.0    | 17    |
| 2025.0 | 12    |
| 1988.0 | 10    |
| 2026.0 | 9     |
| 1986.0 | 8     |
| 1987.0 | 6     |
| 3.0    | 6     |
| 1985.0 | 5     |
| 2031.0 | 5     |
| 2029.0 | 5     |
| 2028.0 | 4     |
| 1982.0 | 4     |

|         |   |
|---------|---|
| 1984.0  | 3 |
| 91.0    | 3 |
| 2.0     | 3 |
| 20165.0 | 2 |
| 1970.0  | 2 |
| 5.0     | 2 |
| 6.0     | 2 |
| 1.0     | 2 |
| 2014.5  | 2 |
| 208.0   | 1 |
| 1979.0  | 1 |
| 38.0    | 1 |
| 2204.0  | 1 |
| 1900.0  | 1 |
| 201.0   | 1 |
| 2011.5  | 1 |
| 83.0    | 1 |
| 4.0     | 1 |
| 1977.0  | 1 |
| 1971.0  | 1 |
| 209.0   | 1 |
| 1976.0  | 1 |
| 2027.0  | 1 |
| 1973.0  | 1 |
| 2106.0  | 1 |
| 1981.0  | 1 |
| 2107.0  | 1 |
| 1972.0  | 1 |
| 2101.0  | 1 |
| 206.0   | 1 |
| 200.0   | 1 |

**dtype:** int64

```
df['orgyear'].value_counts().sort_index(ascending=True)
```





count

orgyear

|        |      |
|--------|------|
| 0.0    | 17   |
| 1.0    | 2    |
| 2.0    | 3    |
| 3.0    | 6    |
| 4.0    | 1    |
| 5.0    | 2    |
| 6.0    | 2    |
| 38.0   | 1    |
| 83.0   | 1    |
| 91.0   | 3    |
| 200.0  | 1    |
| 201.0  | 1    |
| 206.0  | 1    |
| 208.0  | 1    |
| 209.0  | 1    |
| 1900.0 | 1    |
| 1970.0 | 2    |
| 1971.0 | 1    |
| 1972.0 | 1    |
| 1973.0 | 1    |
| 1976.0 | 1    |
| 1977.0 | 1    |
| 1979.0 | 1    |
| 1981.0 | 1    |
| 1982.0 | 4    |
| 1984.0 | 3    |
| 1985.0 | 5    |
| 1986.0 | 8    |
| 1987.0 | 6    |
| 1988.0 | 10   |
| 1989.0 | 22   |
| 1990.0 | 38   |
| 1991.0 | 79   |
| 1992.0 | 47   |
| 1993.0 | 72   |
| 1994.0 | 65   |
| 1995.0 | 94   |
| 1996.0 | 134  |
| 1997.0 | 234  |
| 1998.0 | 279  |
| 1999.0 | 338  |
| 2000.0 | 491  |
| 2001.0 | 708  |
| 2002.0 | 683  |
| 2003.0 | 1010 |
| 2004.0 | 1448 |
| 2005.0 | 1859 |
| 2006.0 | 2066 |

|         |       |
|---------|-------|
| 2007.0  | 2230  |
| 2008.0  | 2688  |
| 2009.0  | 3738  |
| 2010.0  | 5646  |
| 2011.0  | 7802  |
| 2011.5  | 1     |
| 2012.0  | 10257 |
| 2013.0  | 12053 |
| 2014.0  | 16091 |
| 2014.5  | 2     |
| 2015.0  | 19608 |
| 2016.0  | 21476 |
| 2017.0  | 21320 |
| 2018.0  | 22151 |
| 2019.0  | 18545 |
| 2020.0  | 10429 |
| 2021.0  | 3141  |
| 2022.0  | 836   |
| 2023.0  | 245   |
| 2024.0  | 43    |
| 2025.0  | 12    |
| 2026.0  | 9     |
| 2027.0  | 1     |
| 2028.0  | 4     |
| 2029.0  | 5     |
| 2031.0  | 5     |
| 2101.0  | 1     |
| 2106.0  | 1     |
| 2107.0  | 1     |
| 2204.0  | 1     |
| 20165.0 | 2     |

dtype: int64

From value\_counts , can see that there are outliers .

```
df.shape
```

```
(188100, 5)
```

### Outlier Treatment using IQR

```
# Removing outliers from orgyear using IQR.
```

```
q1= df['orgyear'].quantile(0.25)
```

```
q3=df['orgyear'].quantile(0.75)
```

```
iqr=q3-q1
```

```
lower_bound=q1-1.5*iqr
```

```
upper_bound=q3+1.5*iqr
```

```
df=df[(df['orgyear']>=lower_bound) & (df['orgyear']<=upper_bound)]
```

```
df.shape
```

```
(180380, 5)
```

```
df.isna().sum()
```

```
0
company_hash  0
orgyear      0
ctc          0
job_position  0
ctc_updated_year  0
```

```
dtype: int64
```

```
df['ctc'].value_counts().sort_index(ascending=True)
```

```
count
ctc
2      1
6      1
14     1
15     1
16     1
...    ...
199990000  2
200000000  362
250000000  1
255555555  1
1000150000  1
```

```
3296 rows x 1 columns
```

```
dtype: int64
```

```
#Removing outliers from ctc using IQR
```

```
q1= df['ctc'].quantile(0.25)
```

```
q3=df['ctc'].quantile(0.75)
```

```
iqr=q3-q1
```

```
lower_bound=q1-1.5*iqr
```

```
upper_bound=q3+1.5*iqr
```

```
df=df[(df['ctc']>=lower_bound) & (df['ctc']<=upper_bound)]
```

```
df.shape
```

```
(168960, 5)
```

```
df.isna().sum()
```

```
0
company_hash 0
orgyear      0
ctc          0
job_position 0
ctc_updated_year 0
```

dtype: int64

```
df['job_position'].isna().sum()
```

```
0
```

df

```
company_hash  orgyear    ctc  job_position  ctc_updated_year
0      atrgxntxzaxv   2016.0  1100000      other           2020.0
1  qtrxvzwtxzegwgbbrxbxnta  2018.0   449999  fullstackengineer    2019.0
2    ojzwnvwnxwvx   2015.0  2000000  backendengineer    2020.0
3      ngpgutaxv   2017.0   700000  backendengineer    2019.0
4      qxensqghu   2017.0  1400000  fullstackengineer    2019.0
...          ...         ...         ...         ...
205836      mvqwrvo   2011.0  2250000         nan           2019.0
205838      vuurtxzw   2008.0   220000         nan           2019.0
205839      husqvawgb  2017.0   500000         nan           2020.0
205840      vwwgrxnt  2021.0   700000         nan           2021.0
205842      bgqsvzonvzrtj  2014.0  1240000         nan           2016.0
```

168960 rows × 5 columns

Checking for nan strings and converting them to NaN

```
#We see some 'nan's in job_position
df.loc[df['job_position']=='nan','job_position']=np.nan
```

df

```
company_hash  orgyear    ctc  job_position  ctc_updated_year
0      atrgxntxzaxv   2016.0  1100000      other           2020.0
1  qtrxvzwtxzegwgbbrxbxnta  2018.0   449999  fullstackengineer    2019.0
2    ojzwnvwnxwvx   2015.0  2000000  backendengineer    2020.0
3      ngpgutaxv   2017.0   700000  backendengineer    2019.0
4      qxensqghu   2017.0  1400000  fullstackengineer    2019.0
...          ...         ...         ...         ...
205836      mvqwrvo   2011.0  2250000         NaN           2019.0
205838      vuurtxzw   2008.0   220000         NaN           2019.0
205839      husqvawgb  2017.0   500000         NaN           2020.0
205840      vwwgrxnt  2021.0   700000         NaN           2021.0
205842      bgqsvzonvzrtj  2014.0  1240000         NaN           2016.0
```

168960 rows × 5 columns

```
df['job_position'].isna().sum()
```

```
39850
```

```
df['company_hash'].value_counts()
```



|                          | count |
|--------------------------|-------|
| company_hash             |       |
| nvnvwgzohrnvzwjotqcxwto  | 4111  |
| xzegojo                  | 2910  |
| vbvkgz                   | 2227  |
| wgszxkvzn                | 2114  |
| vwwtznht                 | 1998  |
| ...                      | ...   |
| nvrtrznvnngg             | 1     |
| wwwvgogrhnxgzo           | 1     |
| bhzsxtzsxztqxsucnrna     | 1     |
| wvustbxzxmhoxttootqcxwto | 1     |
| wyvqntqwgbbhzwvnxgzo     | 1     |

33384 rows × 1 columns

dtype: int64

Mask companies that have very low occurances.

```
#Masking companies with less count as Others
df['company_hash']=df['company_hash'].mask(df['company_hash'].map(df['company_hash'].value_counts())<5, 'others')
```

```
df['company_hash'].value_counts()
```



|                         | count |
|-------------------------|-------|
| company_hash            |       |
| others                  | 43118 |
| nvnvwgzohrnvzwjotqcxwto | 4111  |
| xzegojo                 | 2910  |
| vbvkgz                  | 2227  |
| wgszxkvzn               | 2114  |
| ...                     | ...   |
| tzuvqvaxsb              | 5     |
| bjbgztjpvqbv            | 5     |
| thqgexzoxnogrhnngxozaxv | 5     |
| wgznxztznrvs            | 5     |
| fxuqg                   | 5     |

3573 rows × 1 columns

dtype: int64

```
df.duplicated().sum()
```



19617

```
df.shape
```



(168960, 5)

```
df.drop_duplicates(inplace=True)
df.shape
```



(149343, 5)

Correcting orgyear values to match current year that are pointing to future years.

```
df[df['orgyear']>=2024].value_counts()
```



|                        |         |         |                       |                  | count |
|------------------------|---------|---------|-----------------------|------------------|-------|
| company_hash           | orgyear | ctc     | job_position          | ctc_updated_year |       |
| hztburgjta             | 2024.0  | 180000  | backendengineer       | 2021.0           | 1     |
| xzntqcxtfmxn           | 2024.0  | 200000  | backendengineer       | 2020.0           | 1     |
| vau                    | 2024.0  | 80000   | other                 | 2020.0           | 1     |
| vbvkgz                 | 2024.0  | 1400000 | frontendengineer      | 2021.0           | 1     |
| xbvonhatzn             | 2024.0  | 110000  | frontendengineer      | 2021.0           | 1     |
|                        |         |         | other                 | 2021.0           | 1     |
|                        | 2025.0  | 300000  | other                 | 2021.0           | 1     |
| xzegojo                | 2025.0  | 360000  | other                 | 2020.0           | 1     |
| xzntqcxtfmxn           | 2024.0  | 400000  | backendengineer       | 2020.0           | 1     |
| ntwybvxyzzaqv          | 2024.0  | 250000  | frontendengineer      | 2021.0           | 1     |
| ywrntwyzgrgsxto        | 2024.0  | 220000  | supportengineer       | 2021.0           | 1     |
| yxptbtootzstq          | 2024.0  | 300000  | androidengineer       | 2021.0           | 1     |
| zgnvuurxwvmrtvwghzn    | 2024.0  | 919999  | engineeringleadership | 2021.0           | 1     |
|                        | 2025.0  | 56000   | backendengineer       | 2021.0           | 1     |
| zgzt                   | 2024.0  | 200000  | other                 | 2021.0           | 1     |
|                        | 2025.0  | 1200000 | fullstackengineer     | 2021.0           | 1     |
| uyvqbtvoj              | 2024.0  | 1800000 | datascientist         | 2019.0           | 1     |
| staxsxnvr              | 2024.0  | 1320000 | backendengineer       | 2020.0           | 1     |
| otretburgjta           | 2024.0  | 200000  | cofounder             | 2021.0           | 1     |
| others                 | 2025.0  | 150000  | devopsengineer        | 2021.0           | 1     |
|                        |         | 120000  | fullstackengineer     | 2018.0           | 1     |
|                        |         | 42000   | frontendengineer      | 2021.0           | 1     |
|                        | 2024.0  | 1400000 | fullstackengineer     | 2019.0           | 1     |
|                        |         | 900000  | other                 | 2021.0           | 1     |
|                        |         | 700000  | fullstackengineer     | 2021.0           | 1     |
|                        |         | 450000  | fullstackengineer     | 2020.0           | 1     |
|                        |         | 220000  | fullstackengineer     | 2021.0           | 1     |
|                        |         | 60000   | frontendengineer      | 2021.0           | 1     |
| onhatznvnbzzxnrvvyvmva | 2024.0  | 1200000 | other                 | 2021.0           | 1     |
|                        |         |         | fullstackengineer     | 2021.0           | 1     |
| onhatzn                | 2025.0  | 1000000 | researchengineers     | 2020.0           | 1     |
|                        | 2024.0  | 100000  | other                 | 2021.0           | 1     |
| nyghsyngqpo            | 2024.0  | 850000  | frontendengineer      | 2021.0           | 1     |
| zvz                    | 2024.0  | 1400000 | engineeringleadership | 2021.0           | 1     |

dtype: int64

```
#changing orgyear above 2024 to 2024
df.loc[df['orgyear']>=2024,'orgyear']=2024
```

```
df[df['orgyear']>2024].value_counts()
```



|              |         |     |              |                  | count |
|--------------|---------|-----|--------------|------------------|-------|
| company_hash | orgyear | ctc | job_position | ctc_updated_year |       |

dtype: int64

Duplicate check and drop them .

```
# checking for duplicates
df.duplicated().sum()
```

 1

```
df.drop_duplicates(inplace=True)
df.shape
```

 (149342, 5)

```
#creating a new feature "years of experience"
```

```
df['years_of_experience']=2024-df['orgyear']
df
```



|        | company_hash            | orgyear | ctc     | job_position      | ctc_updated_year | years_of_experience |  |
|--------|-------------------------|---------|---------|-------------------|------------------|---------------------|-------------------------------------------------------------------------------------|
| 0      | atrgxnntxzaxv           | 2016.0  | 1100000 | other             | 2020.0           | 8.0                 |  |
| 1      | qtrxvzwtxzegwgbbrxbxnta | 2018.0  | 449999  | fullstackengineer | 2019.0           | 6.0                 |  |
| 2      | ojzwnvwnxwvx            | 2015.0  | 2000000 | backendengineer   | 2020.0           | 9.0                 |                                                                                     |
| 3      | ngpgutaxv               | 2017.0  | 700000  | backendengineer   | 2019.0           | 7.0                 |                                                                                     |
| 4      | qxensqghu               | 2017.0  | 1400000 | fullstackengineer | 2019.0           | 7.0                 |                                                                                     |
| ...    | ...                     | ...     | ...     | ...               | ...              | ...                 |                                                                                     |
| 205836 | mvqwrvjjo               | 2011.0  | 2250000 | NaN               | 2019.0           | 13.0                |                                                                                     |
| 205838 | vuurtxzw                | 2008.0  | 220000  | NaN               | 2019.0           | 16.0                |                                                                                     |
| 205839 | husqvawgb               | 2017.0  | 500000  | NaN               | 2020.0           | 7.0                 |                                                                                     |
| 205840 | vwwgrxnt                | 2021.0  | 700000  | NaN               | 2021.0           | 3.0                 |                                                                                     |
| 205842 | bgqsvzonzvrtj           | 2014.0  | 1240000 | NaN               | 2016.0           | 10.0                |                                                                                     |

149342 rows x 6 columns

```
# checking ctc_updated_year feature
```

```
df['ctc_updated_year'].value_counts()
```



|                  | count |
|------------------|-------|
| ctc_updated_year |       |
| 2019.0           | 49095 |
| 2021.0           | 44079 |
| 2020.0           | 36598 |
| 2017.0           | 6488  |
| 2018.0           | 5760  |
| 2016.0           | 4741  |
| 2015.0           | 2581  |

```
dtype: int64
```

```
df[df['ctc_updated_year']<df['orgyear']].value_counts()
```



|               |         |         |                           |                  |                     | count |
|---------------|---------|---------|---------------------------|------------------|---------------------|-------|
| company_hash  | orgyear | ctc     | job_position              | ctc_updated_year | years_of_experience |       |
| adwntwyzgrgsj | 2016.0  | 300000  | fullstackengineer         | 2015.0           | 8.0                 | 1     |
| uyvqbtvoj     | 2024.0  | 1800000 | datascientist             | 2019.0           | 0.0                 | 1     |
| uvjuvr        | 2018.0  | 2000000 | engineeringintern         | 2016.0           | 6.0                 | 1     |
| uvqtdtr       | 2018.0  | 1200000 | other                     | 2016.0           | 6.0                 | 1     |
| uvqvrrtragno  | 2017.0  | 1200000 | datascientist             | 2016.0           | 7.0                 | 1     |
| ...           | ...     | ...     | ...                       | ...              | ...                 | ...   |
| others        | 2017.0  | 60000   | cofounder                 | 2015.0           | 7.0                 | 1     |
|               |         |         | graduateresearchassistant | 2016.0           | 7.0                 | 1     |
|               |         |         | other                     | 2015.0           | 7.0                 | 1     |
|               |         |         |                           | 2016.0           | 7.0                 | 1     |
| zxwtvwnxbxkt  | 2021.0  | 1150000 | datascientist             | 2019.0           | 3.0                 | 1     |

4337 rows × 1 columns

dtype: int64

```
# update the ctc_updated_year with orgyear where ctc_updated_year is less than org year
df.loc[df['ctc_updated_year']<df['orgyear'],'ctc_updated_year']=df['orgyear']
```

```
df[df['ctc_updated_year']<df['orgyear']].value_counts()
```



|              |         |     |              |                  |                     | count |
|--------------|---------|-----|--------------|------------------|---------------------|-------|
| company_hash | orgyear | ctc | job_position | ctc_updated_year | years_of_experience |       |

dtype: int64

```
df.isna().sum()
```



|                     | 0     |
|---------------------|-------|
| company_hash        | 0     |
| orgyear             | 0     |
| ctc                 | 0     |
| job_position        | 34039 |
| ctc_updated_year    | 0     |
| years_of_experience | 0     |

dtype: int64

```
df.loc[df['job_position'].isna()]
```



|        | company_hash             | orgyear | ctc     | job_position | ctc_updated_year | years_of_experience |
|--------|--------------------------|---------|---------|--------------|------------------|---------------------|
| 8      | utqoxontznojontbo        | 2020.0  | 450000  | NaN          | 2020.0           | 4.0                 |
| 9      | others                   | 2019.0  | 360000  | NaN          | 2019.0           | 5.0                 |
| 12     | mvqwrwjowgqugqvntmvzpxzs | 2020.0  | 800000  | NaN          | 2020.0           | 4.0                 |
| 17     | puxn                     | 2020.0  | 1400000 | NaN          | 2020.0           | 4.0                 |
| 18     | mvlvlexzotqc             | 2018.0  | 100000  | NaN          | 2021.0           | 6.0                 |
| ...    | ...                      | ...     | ...     | ...          | ...              | ...                 |
| 205836 | mvqwrvjw                 | 2011.0  | 2250000 | NaN          | 2019.0           | 13.0                |
| 205838 | vuurtxzw                 | 2008.0  | 220000  | NaN          | 2019.0           | 16.0                |
| 205839 | husqvawgb                | 2017.0  | 500000  | NaN          | 2020.0           | 7.0                 |
| 205840 | vwwgrxnt                 | 2021.0  | 700000  | NaN          | 2021.0           | 3.0                 |
| 205842 | bgqsvzonzrtj             | 2014.0  | 1240000 | NaN          | 2016.0           | 10.0                |

34039 rows × 6 columns



```
#update 'job_position' as 'Others' where ever it is null
df.loc[df['job_position'].isna(),'job_position']='others'
```

```
df.shape
```

```
(149342, 6)
```

```
df.isna().sum()
```

```
0
company_hash  0
orgyear       0
ctc           0
job_position  0
ctc_updated_year  0
years_of_experience  0
```

```
dtype: int64
```

```
df.duplicated().sum()
```

```
997
```

```
df.drop_duplicates(inplace=True)
```

```
df.shape
```

```
(148345, 6)
```

```
df.describe()
```

```
orgyear      ctc  ctc_updated_year  years_of_experience
count  148345.000000  1.483450e+05  148345.000000  148345.000000
mean    2015.462479  1.126491e+06    2019.606364    8.537521
std       3.301479  7.420388e+05    1.339959    3.301479
min      2006.000000  2.000000e+00    2015.000000    0.000000
25%      2013.000000  5.700000e+05    2019.000000    6.000000
50%      2016.000000  9.500000e+05    2020.000000    8.000000
75%      2018.000000  1.550000e+06    2021.000000   11.000000
max      2024.000000  3.330000e+06    2024.000000   18.000000
```

Observations:

1. Data is collected from the year 2006 .
2. Max years of experience of the learners is 18 years.

```
# change Others and other to others in job_position
df.loc[df['job_position']=='Others','job_position']='others'
df.loc[df['job_position']=='other','job_position']='others'
```

```
df
```



|        | company_hash             | orgyear | ctc     | job_position      | ctc_updated_year | years_of_experience |   |
|--------|--------------------------|---------|---------|-------------------|------------------|---------------------|-------------------------------------------------------------------------------------|
| 0      | atrgxnntxzaxv            | 2016.0  | 1100000 | others            | 2020.0           | 8.0                 |  |
| 1      | qtrxvzwtxzegwgbbbrxbxnta | 2018.0  | 449999  | fullstackengineer | 2019.0           | 6.0                 |  |
| 2      | ojzwnvwnxwvx             | 2015.0  | 2000000 | backendengineer   | 2020.0           | 9.0                 |                                                                                     |
| 3      | ngpgutaxv                | 2017.0  | 700000  | backendengineer   | 2019.0           | 7.0                 |                                                                                     |
| 4      | qxensqghu                | 2017.0  | 1400000 | fullstackengineer | 2019.0           | 7.0                 |                                                                                     |
| ...    | ...                      | ...     | ...     | ...               | ...              | ...                 |                                                                                     |
| 205836 | mvqwrvj                  | 2011.0  | 2250000 | others            | 2019.0           | 13.0                |                                                                                     |
| 205838 | vuurtxzw                 | 2008.0  | 220000  | others            | 2019.0           | 16.0                |                                                                                     |
| 205839 | husqvawgb                | 2017.0  | 500000  | others            | 2020.0           | 7.0                 |                                                                                     |
| 205840 | vwwgrxnt                 | 2021.0  | 700000  | others            | 2021.0           | 3.0                 |                                                                                     |
| 205842 | bgqsvzonvzrtj            | 2014.0  | 1240000 | others            | 2016.0           | 10.0                |                                                                                     |

148345 rows x 6 columns

```
df.duplicated().sum()
```

 4714

```
pd.set_option('display.max_rows', 1000)
df.head()
```



|   | company_hash             | orgyear | ctc     | job_position      | ctc_updated_year | years_of_experience |  |
|---|--------------------------|---------|---------|-------------------|------------------|---------------------|-------------------------------------------------------------------------------------|
| 0 | atrgxnntxzaxv            | 2016.0  | 1100000 | others            | 2020.0           | 8.0                 |  |
| 1 | qtrxvzwtxzegwgbbbrxbxnta | 2018.0  | 449999  | fullstackengineer | 2019.0           | 6.0                 |                                                                                     |
| 2 | ojzwnvwnxwvx             | 2015.0  | 2000000 | backendengineer   | 2020.0           | 9.0                 |                                                                                     |
| 3 | ngpgutaxv                | 2017.0  | 700000  | backendengineer   | 2019.0           | 7.0                 |                                                                                     |
| 4 | qxensqghu                | 2017.0  | 1400000 | fullstackengineer | 2019.0           | 7.0                 |                                                                                     |

```
df.drop_duplicates(inplace=True)
df.shape
```

 (143631, 6)

## Manual Clustering :

### Manual Clustering Based on Company, Job\_Position and Years\_Of\_experience

```
# group data based on years_of_experience, job_position and company_hash
grouped_df_cjy = df.groupby(['years_of_experience', 'job_position', 'company_hash'])['ctc'].describe()
grouped_df_cjy.head(10)
```



|                     |                       |                     | count | mean      | std           | min       | 25%       | 50%       |
|---------------------|-----------------------|---------------------|-------|-----------|---------------|-----------|-----------|-----------|
| years_of_experience | job_position          | company_hash        |       |           |               |           |           |           |
| 0.0                 | androidengineer       | yxptbtootzstq       | 1.0   | 300000.0  | NaN           | 300000.0  | 300000.0  | 300000.0  |
|                     | backendengineer       | hztburgjta          | 1.0   | 180000.0  | NaN           | 180000.0  | 180000.0  | 180000.0  |
|                     |                       | staxsxnvr           | 1.0   | 1320000.0 | NaN           | 1320000.0 | 1320000.0 | 1320000.0 |
|                     |                       | xzntqcxtfmnx        | 2.0   | 300000.0  | 141421.356237 | 200000.0  | 250000.0  | 300000.0  |
|                     |                       | zgnvuurxwvmrtvwghzn | 1.0   | 56000.0   | NaN           | 56000.0   | 56000.0   | 56000.0   |
|                     | cofounder             | otretburgjta        | 1.0   | 200000.0  | NaN           | 200000.0  | 200000.0  | 200000.0  |
|                     | datascientist         | uyvqbtvoj           | 1.0   | 1800000.0 | NaN           | 1800000.0 | 1800000.0 | 1800000.0 |
|                     | devopsengineer        | others              | 1.0   | 150000.0  | NaN           | 150000.0  | 150000.0  | 150000.0  |
|                     | engineeringleadership | zgnvuurxwvmrtvwghzn | 1.0   | 919999.0  | NaN           | 919999.0  | 919999.0  | 919999.0  |
|                     |                       | zvz                 | 1.0   | 1400000.0 | NaN           | 1400000.0 | 1400000.0 | 1400000.0 |

Next steps:

[Generate code with grouped\\_df\\_cjy](#)[View recommended plots](#)[New interactive sheet](#)

```
df_cjy= df.merge(grouped_df_cjy, on=['years_of_experience', 'job_position', 'company_hash'],how='left')
df_cjy.head()
```

|   | company_hash           | orgyear | ctc     | job_position      | ctc_updated_year | years_of_experience | count | mean                  |
|---|------------------------|---------|---------|-------------------|------------------|---------------------|-------|-----------------------|
| 0 | atrgxnntxzaxv          | 2016.0  | 1100000 | others            | 2020.0           | 8.0                 | 1.0   | 1.100000e+06          |
| 1 | qtrxvzwtzegwgbbrxbxnta | 2018.0  | 449999  | fullstackengineer | 2019.0           | 6.0                 | 7.0   | 7.742856e+05 250922.3 |
| 2 | ojzwnvwnxwvx           | 2015.0  | 2000000 | backendengineer   | 2020.0           | 9.0                 | 1.0   | 2.000000e+06          |
| 3 | ngpgutaxv              | 2017.0  | 700000  | backendengineer   | 2019.0           | 7.0                 | 7.0   | 1.158571e+06 404780.9 |
| 4 | qxensqghu              | 2017.0  | 1400000 | fullstackengineer | 2019.0           | 7.0                 | 1.0   | 1.400000e+06          |

Creating designation based on the salary they are getting in their respective company

```
def segment(a,b_50,c_75):
    if a<=b_50:
        return 3
    elif a>b_50 and a<=c_75:
        return 2
    else:
        return 1
```

```
df.head()
```

|   | company_hash           | orgyear | ctc     | job_position      | ctc_updated_year | years_of_experience |
|---|------------------------|---------|---------|-------------------|------------------|---------------------|
| 0 | atrgxnntxzaxv          | 2016.0  | 1100000 | others            | 2020.0           | 8.0                 |
| 1 | qtrxvzwtzegwgbbrxbxnta | 2018.0  | 449999  | fullstackengineer | 2019.0           | 6.0                 |
| 2 | ojzwnvwnxwvx           | 2015.0  | 2000000 | backendengineer   | 2020.0           | 9.0                 |
| 3 | ngpgutaxv              | 2017.0  | 700000  | backendengineer   | 2019.0           | 7.0                 |
| 4 | qxensqghu              | 2017.0  | 1400000 | fullstackengineer | 2019.0           | 7.0                 |

```
df_cjy['designation']=df_cjy.apply(lambda x: segment(x['ctc'],x['50%'],x['75%']),axis=1)
df_cjy.head()
```

|   | company_hash           | orgyear | ctc     | job_position      | ctc_updated_year | years_of_experience | count | mean                  |
|---|------------------------|---------|---------|-------------------|------------------|---------------------|-------|-----------------------|
| 0 | atrgxnntxzaxv          | 2016.0  | 1100000 | others            | 2020.0           | 8.0                 | 1.0   | 1.100000e+06          |
| 1 | qtrxvzwtzegwgbbrxbxnta | 2018.0  | 449999  | fullstackengineer | 2019.0           | 6.0                 | 7.0   | 7.742856e+05 250922.3 |
| 2 | ojzwnvwnxwvx           | 2015.0  | 2000000 | backendengineer   | 2020.0           | 9.0                 | 1.0   | 2.000000e+06          |
| 3 | ngpgutaxv              | 2017.0  | 700000  | backendengineer   | 2019.0           | 7.0                 | 7.0   | 1.158571e+06 404780.9 |
| 4 | qxensqghu              | 2017.0  | 1400000 | fullstackengineer | 2019.0           | 7.0                 | 1.0   | 1.400000e+06          |

```
df_cjy['designation'].value_counts(normalize=True)*100
```

|             | proportion |
|-------------|------------|
| designation |            |
| 3           | 66.042846  |
| 1           | 21.619985  |
| 2           | 12.337170  |

```
dtype: float64
```

## Manual Clustering Based on Company and Job\_Position

```
# group data based on job_position and company_hash
grouped_df_cj=df.groupby(['job_position','company_hash'])['ctc'].describe()
grouped_df_cj.head(10)
```

|                          |                      | count | mean      | std | min       | 25%       | 50%       | 75%       | max       |
|--------------------------|----------------------|-------|-----------|-----|-----------|-----------|-----------|-----------|-----------|
| job_position             | company_hash         |       |           |     |           |           |           |           |           |
| abapdeveloper            | others               | 1.0   | 500000.0  | NaN | 500000.0  | 500000.0  | 500000.0  | 500000.0  | 500000.0  |
| account                  | zgnvuurxwvmrtvwwghzn | 1.0   | 2000000.0 | NaN | 2000000.0 | 2000000.0 | 2000000.0 | 2000000.0 | 2000000.0 |
| administrativeclerk      | others               | 1.0   | 500000.0  | NaN | 500000.0  | 500000.0  | 500000.0  | 500000.0  | 500000.0  |
| administrator            | others               | 1.0   | 380000.0  | NaN | 380000.0  | 380000.0  | 380000.0  | 380000.0  | 380000.0  |
| advisor                  | exo                  | 1.0   | 360000.0  | NaN | 360000.0  | 360000.0  | 360000.0  | 360000.0  | 360000.0  |
| advisorysoftwareengineer | others               | 1.0   | 1290000.0 | NaN | 1290000.0 | 1290000.0 | 1290000.0 | 1290000.0 | 1290000.0 |
| advisorysystemanalyt     | xmb                  | 1.0   | 680000.0  | NaN | 680000.0  | 680000.0  | 680000.0  | 680000.0  | 680000.0  |
| agencycollectionmanager  | others               | 1.0   | 360000.0  | NaN | 360000.0  | 360000.0  | 360000.0  | 360000.0  | 360000.0  |
| agroupchatapplication    | others               | 1.0   | 500000.0  | NaN | 500000.0  | 500000.0  | 500000.0  | 500000.0  | 500000.0  |
| aiengineer               | others               | 1.0   | 1000000.0 | NaN | 1000000.0 | 1000000.0 | 1000000.0 | 1000000.0 | 1000000.0 |

Next steps: [Generate code with grouped\\_df\\_cj](#) [View recommended plots](#) [New interactive sheet](#)

Merging the grouped\_cj with df based on company and job\_postion

```
df_cj=df.merge(grouped_df_cj,on=['job_position','company_hash'],how='left')
df_cj.head()
```

|   | company_hash            | orgyear | ctc     | job_position      | ctc_updated_year | years_of_experience | count | mean         |
|---|-------------------------|---------|---------|-------------------|------------------|---------------------|-------|--------------|
| 0 | atrgxnntxzaxv           | 2016.0  | 1100000 | others            | 2020.0           | 8.0                 | 2.0   | 1.085000e+06 |
| 1 | qtrxvzwtxzegwgbbrxbxnta | 2018.0  | 449999  | fullstackengineer | 2019.0           | 6.0                 | 25.0  | 9.882000e+05 |
| 2 | ojzwnvwnxwvx            | 2015.0  | 2000000 | backendengineer   | 2020.0           | 9.0                 | 3.0   | 1.983333e+06 |
| 3 | ngpgutaxv               | 2017.0  | 700000  | backendengineer   | 2019.0           | 7.0                 | 24.0  | 1.416667e+06 |
| 4 | qxensqghu               | 2017.0  | 1400000 | fullstackengineer | 2019.0           | 7.0                 | 3.0   | 8.466667e+05 |

Creating class based on the salary they are getting in their respective company

```
df_cj['class']=df_cj.apply(lambda x: segment(x['ctc'],x['50%'],x['75%']),axis=1)
df_cj.head()
```

|   | company_hash            | orgyear | ctc     | job_position      | ctc_updated_year | years_of_experience | count | mean         |
|---|-------------------------|---------|---------|-------------------|------------------|---------------------|-------|--------------|
| 0 | atrgxnntxzaxv           | 2016.0  | 1100000 | others            | 2020.0           | 8.0                 | 2.0   | 1.085000e+06 |
| 1 | qtrxvzwtxzegwgbbrxbxnta | 2018.0  | 449999  | fullstackengineer | 2019.0           | 6.0                 | 25.0  | 9.882000e+05 |
| 2 | ojzwnvwnxwvx            | 2015.0  | 2000000 | backendengineer   | 2020.0           | 9.0                 | 3.0   | 1.983333e+06 |
| 3 | ngpgutaxv               | 2017.0  | 700000  | backendengineer   | 2019.0           | 7.0                 | 24.0  | 1.416667e+06 |
| 4 | qxensqghu               | 2017.0  | 1400000 | fullstackengineer | 2019.0           | 7.0                 | 3.0   | 8.466667e+05 |

```
df_cj['class'].value_counts(normalize=True)*100
```

| class | proportion |
|-------|------------|
| 3     | 56.401473  |
| 1     | 24.372872  |
| 2     | 19.225655  |

dtype: float64

```
# job_position that has highest_class
df_cj[df_cj['class']==1][['job_position','ctc']].groupby('job_position')['ctc'].describe().sort_values(by=['count','mean'],a
```



|                       | count   | mean         | std           | min      | 25%       | 50%       | 75%       | max       |
|-----------------------|---------|--------------|---------------|----------|-----------|-----------|-----------|-----------|
| job_position          |         |              |               |          |           |           |           |           |
| others                | 10408.0 | 1.897610e+06 | 702854.115232 | 50000.0  | 1370000.0 | 1850000.0 | 2500000.0 | 3330000.0 |
| backendengineer       | 8275.0  | 2.048395e+06 | 693553.460515 | 12000.0  | 1500000.0 | 2000000.0 | 2600000.0 | 3330000.0 |
| fullstackengineer     | 4836.0  | 1.826031e+06 | 692341.866726 | 59000.0  | 1300000.0 | 1720000.0 | 2330000.0 | 3330000.0 |
| frontendengineer      | 2015.0  | 1.655688e+06 | 632196.773290 | 90000.0  | 1200000.0 | 1550000.0 | 2000000.0 | 3300000.0 |
| qaengineer            | 1268.0  | 1.384218e+06 | 631517.411210 | 65000.0  | 900000.0  | 1250000.0 | 1700000.0 | 3300000.0 |
| datascientist         | 1028.0  | 1.985017e+06 | 660657.952259 | 115000.0 | 1600000.0 | 2000000.0 | 2452500.0 | 3310000.0 |
| sdet                  | 997.0   | 1.537592e+06 | 718557.169574 | 140000.0 | 900000.0  | 1400000.0 | 2000000.0 | 3329999.0 |
| devopsengineer        | 929.0   | 1.740286e+06 | 629445.066247 | 56000.0  | 1300000.0 | 1650000.0 | 2100000.0 | 3300000.0 |
| androidengineer       | 919.0   | 1.765926e+06 | 649457.401094 | 14000.0  | 1300000.0 | 1664000.0 | 2200000.0 | 3300000.0 |
| engineeringleadership | 788.0   | 2.531369e+06 | 589970.478686 | 124000.0 | 2225000.0 | 2640000.0 | 3000000.0 | 3320000.0 |



Manual Clustering based on company

```
# group data based on company
grouped_df_c=df.groupby(['company_hash'])['ctc'].describe()
grouped_df_c.head(10)
```



|                                    | count | mean         | std           | min       | 25%       | 50%       | 75%       | max       |
|------------------------------------|-------|--------------|---------------|-----------|-----------|-----------|-----------|-----------|
| company_hash                       |       |              |               |           |           |           |           |           |
| aaqxctzavnvowxtzwtovzvrjnxwoucnrna | 7.0   | 6.114286e+05 | 362373.130508 | 360000.0  | 430000.0  | 500000.0  | 580000.0  | 1400000.0 |
| abwavnvojontb                      | 5.0   | 7.320000e+05 | 43817.804600  | 700000.0  | 700000.0  | 700000.0  | 780000.0  | 780000.0  |
| adwntwyzgrrsj                      | 245.0 | 7.420204e+05 | 510138.354048 | 32000.0   | 400000.0  | 570000.0  | 930000.0  | 3200000.0 |
| adwntwyzgrrsxto                    | 126.0 | 7.016111e+05 | 376552.552020 | 100000.0  | 401750.0  | 600000.0  | 900000.0  | 2350000.0 |
| agdutq                             | 6.0   | 1.666667e+06 | 686051.504383 | 800000.0  | 1400000.0 | 1400000.0 | 2225000.0 | 2500000.0 |
| aggartmrhtxzzgcvnxgzo              | 9.0   | 5.511111e+05 | 212805.336190 | 360000.0  | 450000.0  | 500000.0  | 540000.0  | 1000000.0 |
| aghmznzhn                          | 18.0  | 1.221667e+06 | 678883.165899 | 500000.0  | 701250.0  | 875000.0  | 1800000.0 | 2400000.0 |
| agnoihvqto                         | 19.0  | 7.230526e+05 | 318784.753867 | 340000.0  | 485000.0  | 700000.0  | 900000.0  | 1600000.0 |
| agnut                              | 8.0   | 1.653750e+06 | 574927.014001 | 1230000.0 | 1350000.0 | 1425000.0 | 1687500.0 | 3000000.0 |
| agnutxz                            | 5.0   | 1.384000e+06 | 682114.359913 | 700000.0  | 800000.0  | 1220000.0 | 2100000.0 | 2100000.0 |



Next steps:

[Generate code with grouped\\_df\\_c](#)

[View recommended plots](#)

[New interactive sheet](#)

Merge group\_df\_c with df

```
df_c=df.merge(grouped_df_c,on=['company_hash'],how='left')
df_c.head()
```



|   | company_hash            | orgyear | ctc     | job_position      | ctc_updated_year | years_of_experience | count | mean                  |
|---|-------------------------|---------|---------|-------------------|------------------|---------------------|-------|-----------------------|
| 0 | atrgxnntxzaxv           | 2016.0  | 1100000 | others            | 2020.0           | 8.0                 | 9.0   | 1.115667e+06 458111.8 |
| 1 | qtrxvzwtxzegwgbbrxbxnta | 2018.0  | 449999  | fullstackengineer | 2019.0           | 6.0                 | 377.0 | 1.065469e+06 636791.1 |
| 2 | ojzwnvwnxwvx            | 2015.0  | 2000000 | backendengineer   | 2020.0           | 9.0                 | 5.0   | 1.860000e+06 711863.7 |
| 3 | ngpgutaxv               | 2017.0  | 700000  | backendengineer   | 2019.0           | 7.0                 | 59.0  | 1.455508e+06 655423.4 |
| 4 | qxensqghu               | 2017.0  | 1400000 | fullstackengineer | 2019.0           | 7.0                 | 6.0   | 9.400000e+05 389871.7 |

```
df_c.duplicated().sum()
```



0

Create Tier based on the company

```
df_c['tier']=df_c.apply(lambda x: segment(x['ctc'],x['50%'],x['75%']),axis=1)
```

```
df_c.head()
```

|   | company_hash            | orgyear | ctc     | job_position      | ctc_updated_year | years_of_experience | count | mean         |
|---|-------------------------|---------|---------|-------------------|------------------|---------------------|-------|--------------|
| 0 | atrgxnntxzaxv           | 2016.0  | 1100000 | others            | 2020.0           | 8.0                 | 9.0   | 1.115667e+06 |
| 1 | qtrxvzwtxzegwgbbrxbxnta | 2018.0  | 449999  | fullstackengineer | 2019.0           | 6.0                 | 377.0 | 1.065469e+06 |
| 2 | ojzwnvwnxwvx            | 2015.0  | 2000000 | backendengineer   | 2020.0           | 9.0                 | 5.0   | 1.860000e+06 |
| 3 | ngpgutaxv               | 2017.0  | 700000  | backendengineer   | 2019.0           | 7.0                 | 59.0  | 1.455508e+06 |
| 4 | qxensqghu               | 2017.0  | 1400000 | fullstackengineer | 2019.0           | 7.0                 | 6.0   | 9.400000e+05 |

```
df_c['tier'].value_counts(normalize=True)*100
```

| proportion |           |
|------------|-----------|
| tier       |           |
| 3          | 52.710766 |
| 1          | 23.675947 |
| 2          | 23.613287 |

dtype: float64

```
# drop stats columns from df_cjy,df_cj,df_c
df_cjy.drop(columns=['count','mean','std','min','25%','50%','75%','max'],axis=1,inplace=True)
df_cj.drop(columns=['count','mean','std','min','25%','50%','75%','max'],axis=1,inplace=True)
#df_c.drop(columns=['count','mean','std','min','25%','50%','75%','max'],axis=1,inplace=True)
```

```
df_cjy.head()
```

|   | company_hash            | orgyear | ctc     | job_position      | ctc_updated_year | years_of_experience | designation |
|---|-------------------------|---------|---------|-------------------|------------------|---------------------|-------------|
| 0 | atrgxnntxzaxv           | 2016.0  | 1100000 | others            | 2020.0           | 8.0                 | 3           |
| 1 | qtrxvzwtxzegwgbbrxbxnta | 2018.0  | 449999  | fullstackengineer | 2019.0           | 6.0                 | 3           |
| 2 | ojzwnvwnxwvx            | 2015.0  | 2000000 | backendengineer   | 2020.0           | 9.0                 | 3           |
| 3 | ngpgutaxv               | 2017.0  | 700000  | backendengineer   | 2019.0           | 7.0                 | 3           |
| 4 | qxensqghu               | 2017.0  | 1400000 | fullstackengineer | 2019.0           | 7.0                 | 3           |

```
#merge df_cjy,df_cj,df_c on company_hash, orgyear, ctc, job_position,ctc_updated_year,years_of_experience
df_final=df_cjy.merge(df_cj,on=['company_hash','orgyear','ctc','job_position','ctc_updated_year','years_of_experience'],how=
```

```
df_final.head()
```

|   | company_hash            | orgyear | ctc     | job_position      | ctc_updated_year | years_of_experience | designation | class |
|---|-------------------------|---------|---------|-------------------|------------------|---------------------|-------------|-------|
| 0 | atrgxnntxzaxv           | 2016.0  | 1100000 | others            | 2020.0           | 8.0                 | 3           | 1     |
| 1 | qtrxvzwtxzegwgbbrxbxnta | 2018.0  | 449999  | fullstackengineer | 2019.0           | 6.0                 | 3           | 3     |
| 2 | ojzwnvwnxwvx            | 2015.0  | 2000000 | backendengineer   | 2020.0           | 9.0                 | 3           | 3     |
| 3 | ngpgutaxv               | 2017.0  | 700000  | backendengineer   | 2019.0           | 7.0                 | 3           | 3     |
| 4 | qxensqghu               | 2017.0  | 1400000 | fullstackengineer | 2019.0           | 7.0                 | 3           | 1     |

```
# merge df_final with df_c on comapany_hash
data=df_final.merge(df_c,on=['company_hash','orgyear','ctc','job_position','ctc_updated_year','years_of_experience'],how='le
```

data

|        | company_hash            | orgyear | ctc     | job_position      | ctc_updated_year | years_of_experience | designation | class | coi |
|--------|-------------------------|---------|---------|-------------------|------------------|---------------------|-------------|-------|-----|
| 0      | atrgxnntxzaxv           | 2016.0  | 1100000 | others            | 2020.0           | 8.0                 | 3           | 1     |     |
| 1      | qtrxvzwtxzegwgbbrxbxnta | 2018.0  | 449999  | fullstackengineer | 2019.0           | 6.0                 | 3           | 3     | 37  |
| 2      | ojzwnvwnxwvx            | 2015.0  | 2000000 | backendengineer   | 2020.0           | 9.0                 | 3           | 3     |     |
| 3      | ngpgutaxv               | 2017.0  | 700000  | backendengineer   | 2019.0           | 7.0                 | 3           | 3     | 5   |
| 4      | qxensqghu               | 2017.0  | 1400000 | fullstackengineer | 2019.0           | 7.0                 | 3           | 1     |     |
| ...    | ...                     | ...     | ...     | ...               | ...              | ...                 | ...         | ...   | ... |
| 143626 | mvqwrwjo                | 2011.0  | 2250000 | others            | 2019.0           | 13.0                | 1           | 1     | 6   |
| 143627 | vuurtxzw                | 2008.0  | 220000  | others            | 2019.0           | 16.0                | 3           | 3     | 1   |
| 143628 | husqvawgb               | 2017.0  | 500000  | others            | 2020.0           | 7.0                 | 3           | 3     | 5   |
| 143629 | vwwgrxnt                | 2021.0  | 700000  | others            | 2021.0           | 3.0                 | 3           | 3     | 15  |
| 143630 | bgqsvzonzvrtj           | 2014.0  | 1240000 | others            | 2016.0           | 10.0                | 3           | 3     | 46  |

143631 rows x 17 columns

*\*Questions based on Manual Clustering \**

1) Top 10 employees (earning more than most of the employees in the company) - Tier 1 \* Top 10 employees of data science in each company earning more than their peers - Class 1 \* Bottom 10 employees of data science in each company earning less than their peers - Class 3

**Top 10 employees (earning more than most of the employees in the company) - Tier 1**

```
#Top 10 employees (earning more than most of the employees in the company) - Tier 1
data[data['tier']==1][['company_hash','job_position','ctc']].sort_values(by='ctc',ascending=False).head(10)
```

|        | company_hash           | job_position          | ctc     |
|--------|------------------------|-----------------------|---------|
| 95146  | bxwqgogenqtotvqwyxzaxv | researchengineers     | 3330000 |
| 133098 | btaxvpaxa              | fullstackengineer     | 3330000 |
| 121227 | btaxvpaxa              | backendengineer       | 3330000 |
| 84509  | uvjovetsqghu           | backendengineer       | 3330000 |
| 122185 | bxwqgogenqtotvqwyxzaxv | others                | 3330000 |
| 40787  | nvnvwrxi               | productdesigner       | 3329999 |
| 94119  | erxupvqn               | backendengineer       | 3329999 |
| 132299 | others                 | backendarchitect      | 3329999 |
| 97711  | mtwbpvzwghrntq         | engineeringleadership | 3329999 |
| 48567  | uvjnb                  | sdet                  | 3329999 |

```
# rows with job_position datascience
data[data['job_position'].str.startswith('d')].value_counts()
```



| company_hash  | orgyear | ctc     | job_position   | ctc_updated_year | years_of_experience | designation | class | count   | me:        |
|---------------|---------|---------|----------------|------------------|---------------------|-------------|-------|---------|------------|
| adwntwyzgrgsj | 2006.0  | 1200000 | devopsengineer | 2019.0           | 18.0                | 3           | 1     | 245.0   | 7.420204e+ |
| txxwoogz      | 2011.0  | 1050000 | devopsengineer | 2019.0           | 13.0                | 3           | 2     | 307.0   | 1.001998e+ |
|               | 2008.0  | 1370000 | devopsengineer | 2019.0           | 16.0                | 3           | 1     | 307.0   | 1.001998e+ |
|               |         | 2400000 | datascientist  | 2019.0           | 16.0                | 3           | 1     | 307.0   | 1.001998e+ |
|               | 2009.0  | 1000000 | datascientist  | 2019.0           | 15.0                | 3           | 3     | 307.0   | 1.001998e+ |
| ...           | ...     | ...     | ...            | ...              | ...                 | ...         | ...   | ...     | ...        |
| others        | 2006.0  | 1500000 | dataanalyst    | 2016.0           | 18.0                | 2           | 1     | 22086.0 | 9.606033e+ |
|               |         |         | devopsengineer | 2019.0           | 18.0                | 3           | 1     | 22086.0 | 9.606033e+ |
|               |         | 1600000 | datascientist  | 2019.0           | 18.0                | 3           | 1     | 22086.0 | 9.606033e+ |
|               |         | 1650000 | devopsengineer | 2019.0           | 18.0                | 2           | 1     | 22086.0 | 9.606033e+ |
| zxztrtvuo     | 2019.0  | 1250000 | datascientist  | 2021.0           | 5.0                 | 3           | 3     | 66.0    | 9.802727e+ |

11059 rows x 10 columns

dtype: int64

2)Bottom 10 employees (earning less than most of the employees in the company)- Tier 3

```
#Top 10 employees of data science in each company earning more than their peers - Class 1
df_class1=data[(data['job_position']=='datascientist')&(data['class']==1)][['company_hash','ctc','class','job_position']].sc
df_class1.head(10)
```



|        | company_hash      | ctc     | class | job_position  |
|--------|-------------------|---------|-------|---------------|
| 141059 | others            | 3310000 | 1     | datascientist |
| 81870  | wxnx              | 3300000 | 1     | datascientist |
| 95213  | st                | 3300000 | 1     | datascientist |
| 41406  | fzosrgmvrotqcxwto | 3300000 | 1     | datascientist |
| 11478  | bxwqgogen         | 3300000 | 1     | datascientist |
| 63686  | bxwqgogen         | 3300000 | 1     | datascientist |
| 65895  | bxwqgogen         | 3300000 | 1     | datascientist |
| 88812  | stytrnywvqt       | 3300000 | 1     | datascientist |
| 94671  | lxg               | 3300000 | 1     | datascientist |
| 85698  | others            | 3300000 | 1     | datascientist |



Next steps:

[Generate code with df\\_class1](#)

[View recommended plots](#)

[New interactive sheet](#)

Top 10 employees in each company - X department - having 5/6/7 years of experience earning more than their peers - Tier X

```
#Bottom 10 employees of data science in each company earning less than their peers - Class 3
df_class3=data[(data['job_position']=='datascientist')&(data['class']==3)][['company_hash','ctc','class','job_position']].sc
df_class3.head(10)
```





|        | company_hash                  | ctc  | class | job_position  |                                                                                   |
|--------|-------------------------------|------|-------|---------------|-----------------------------------------------------------------------------------|
| 9032   | srgmvrstaxzntrrxstzwtgenyxzso | 4000 | 3     | datascientist |  |
| 118586 | others                        | 4000 | 3     | datascientist |                                                                                   |
| 7349   | bxyhuwgbbhzxwvnxgz            | 4000 | 3     | datascientist |                                                                                   |
| 136047 | others                        | 4000 | 3     | datascientist |                                                                                   |
| 38081  | onhatzn                       | 6000 | 3     | datascientist |                                                                                   |
| 97633  | ovbohzstrtnqgbtwyvxwo         | 7000 | 3     | datascientist |                                                                                   |
| 18861  | others                        | 7200 | 3     | datascientist |                                                                                   |
| 128677 | others                        | 7250 | 3     | datascientist |                                                                                   |
| 67709  | others                        | 7500 | 3     | datascientist |                                                                                   |
| 7898   | nvnvwgzohrnrvzwjotqcxwto      | 7500 | 3     | datascientist |                                                                                   |

Next steps: [Generate code with df\\_class3](#) [View recommended plots](#) [New interactive sheet](#)

Bottom 10 employees (earning less than most of the employees in the company)- Tier 3

```
#Bottom 10 employees (earning less than most of the employees in the company)- Tier 3
data[data['tier']==3][['company_hash','job_position','ctc']].sort_values(by='ctc',ascending=True).head(10)
```



|        | company_hash | job_position          | ctc |                                                                                   |
|--------|--------------|-----------------------|-----|-----------------------------------------------------------------------------------|
| 96538  | xzntqcxftmxn | backendengineer       | 2   |  |
| 84757  | xzntqcxftmxn | others                | 6   |                                                                                   |
| 81917  | xzntqcxftmxn | others                | 14  |                                                                                   |
| 129869 | others       | others                | 15  |                                                                                   |
| 129073 | others       | others                | 16  |                                                                                   |
| 41073  | others       | others                | 24  |                                                                                   |
| 66947  | others       | androidengineer       | 25  |                                                                                   |
| 83920  | others       | others                | 200 |                                                                                   |
| 117277 | others       | databaseadministrator | 300 |                                                                                   |
| 60119  | others       | cofounder             | 500 |                                                                                   |

```
#Top 10 employees in each company - X department - having 5/6/7 years of experience earning more than their peers - Tier X
df_tierX=data.groupby(['tier','company_hash'])[['years_of_experience','company_hash','job_position','tier']]
```

df\_tierX

```
<pandas.core.groupby.generic.DataFrameGroupBy object at 0x7caaa987eb30>
```

```
df_tierX_filtered = data.groupby(['tier', 'company_hash', 'job_position']).filter(
    lambda x: x['years_of_experience'].isin([5, 6, 7]).any()
).sort_index()
```

```
df_tierX_final=df_tierX_filtered[['tier','company_hash','job_position','years_of_experience']]
```

```
df_tierX_filtered = data[data['years_of_experience'].isin([5, 6, 7])].sort_index()
```

```
df_tierX_filtered.head()
```



|   | company_hash                    | orgyear | ctc     | job_position      | ctc_updated_year | years_of_experience | designation | class | c |
|---|---------------------------------|---------|---------|-------------------|------------------|---------------------|-------------|-------|---|
| 1 | qtrxvzwtzegwgbbrxbxnta          | 2018.0  | 449999  | fullstackengineer | 2019.0           | 6.0                 | 3           | 3     |   |
| 3 | ngpgutaxv                       | 2017.0  | 700000  | backendengineer   | 2019.0           | 7.0                 | 3           | 3     |   |
| 4 | qxensqghu                       | 2017.0  | 1400000 | fullstackengineer | 2019.0           | 7.0                 | 3           | 1     |   |
| 5 | yvuuxrjhzbvqqxtabvqptnxzsucnrna | 2018.0  | 700000  | fullstackengineer | 2020.0           | 6.0                 | 3           | 3     |   |
| 6 | lubgqsvzwvyvotwg                | 2018.0  | 1500000 | fullstackengineer | 2019.0           | 6.0                 | 3           | 3     |   |

Next steps:

Generate code with df\_tierX\_filtered

View recommended plots

New interactive sheet

```
df_tierX_final = df_tierX_filtered.groupby(['tier', 'company_hash', 'job_position'])[['company_hash','job_position','years_c
df_tierX_final.head()
```

|   | company_hash                     | job_position          | years_of_experience | tier |  |
|---|----------------------------------|-----------------------|---------------------|------|--|
| 0 | aaqxctzavnvowxtzwtovzvrjnxwoucna | backendengineer       | 6.0                 | 1    |  |
| 1 | adwntwyzgrgsj                    | datascientist         | 7.0                 | 1    |  |
| 2 | adwntwyzgrgsj                    | noncoder              | 7.0                 | 1    |  |
| 3 | adwntwyzgrgsj                    | others                | 7.0                 | 1    |  |
| 4 | adwntwyzgrgsxto                  | databaseadministrator | 7.0                 | 1    |  |

Next steps:

Generate code with df\_tierX\_final

View recommended plots

New interactive sheet

Top 10 employees in each company - X department - having 5/6/7 years of experience earning more than their peers - Tier X

```
df_tierX_final.groupby('company_hash')[['job_position','years_of_experience']].value_counts().sort_values(ascending=True).he
```

|  | company_hash           | job_position       | years_of_experience | count |
|--|------------------------|--------------------|---------------------|-------|
|  | zxztrtvuo              | fullstackengineer  | 7.0                 | 1     |
|  | otwhqbtntqorxbxnta     | backendengineer    | 7.0                 | 1     |
|  |                        | dataanalyst        | 5.0                 | 1     |
|  |                        | others             | 5.0                 | 1     |
|  | otwhqxex               | engineeringintern  | 6.0                 | 1     |
|  | wgxznqxmntntwyzgrgsxto | programmanager     | 6.0                 | 1     |
|  | otzaxzmrht             | backendengineer    | 7.0                 | 1     |
|  |                        | frontendengineer   | 7.0                 | 1     |
|  | otwhqgzxd              | securityleadership | 6.0                 | 1     |
|  | otzaxzmrht             | others             | 7.0                 | 1     |

dtype: int64

Top 10 companies (based on their CTC)

```
#Top 10 companies (based on their CTC)
data.groupby('company_hash')['ctc'].mean().sort_values(ascending=False).head(10)
```

|  | company_hash            | ctc          |
|--|-------------------------|--------------|
|  | yvqztooxg               | 2.886667e+06 |
|  | yxsyutvp                | 2.800000e+06 |
|  | mgongzgwzohrnzssqghumws | 2.788000e+06 |
|  | gmotqctvxkrvmo          | 2.771429e+06 |
|  | wrhonq                  | 2.728000e+06 |
|  | wgatzvnxgz              | 2.684444e+06 |
|  | vuuajzvbwo              | 2.625000e+06 |
|  | vwvtrtnqvat             | 2.616667e+06 |
|  | wgzcxcv                 | 2.568750e+06 |
|  | pyvnmvggp               | 2.560000e+06 |

dtype: float64

Get top 2 positions in every company based on CTC

```
# Get top 2 positions in every company based on CTC
```

```
# Get top 2 positions in every company based on CTC
```

```
top_positions = data.groupby('company_hash')[['job_position', 'ctc']].apply(lambda x: x.nlargest(2, 'ctc')).reset_index(lev
print(top_positions)
```

```
↗
company_hash      job_position      ctc
aaqxctzavnvowxtzwtovzvrjnxwoucnrna  backendengineer  1400000
aaqxctzavnvowxtzwtovzvrjnxwoucnrna  backendengineer  6600000
abwavnvojontb      backendengineer  7800000
abwavnvojontb      fullstackengineer  7800000
adwntwyzgrgsj      devopsengineer   3200000
...
zxyxrtznntwyzgrgsxto  backendengineer  1300000
zxzlvwvqn             others         3000000
zxzlvwvqn             backendengineer  3000000
zxztrtvuo             devopsengineer  2700000
zxztrtvuo             frontendengineer  2500000
```

```
[7146 rows x 2 columns]
```

## ✓ Unsupervised Learning

- Checking clustering tendency
- Elbow method
- K-means clustering
- Hierarchical clustering (you can do this on a sample of the dataset if your process is taking time)

```
# Import libraries
```

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
from sklearn.preprocessing import LabelEncoder,MinMaxScaler
```

```
from sklearn.cluster import KMeans
```

```
from sklearn.metrics import silhouette_score
```

```
from sklearn.cluster import AgglomerativeClustering
```

```
X = data[['company_hash','job_position','ctc', 'years_of_experience', 'class', 'designation', 'tier']].copy()
X
```

```
↗
company_hash  job_position  ctc  years_of_experience  class  designation  tier
0      atrgxntxzaxv      others  1100000              8.0      1              3      2
1      qtrxvzwtxzegwgbbrxbxnta  fullstackengineer  449999              6.0      3              3      3
2      ojzwnvwnxwvx  backendengineer  2000000              9.0      3              3      3
3      ngpgutaxv  backendengineer  700000              7.0      3              3      3
4      qxensqghu  fullstackengineer  1400000              7.0      1              3      1
...
143626      mvqwrvj0      others  2250000             13.0      1              1      1
143627      vuurtxzw      others  220000             16.0      3              3      3
143628      husqvawgb      others  500000              7.0      3              3      3
143629      vwwgrxnt      others  700000              3.0      3              3      3
143630      bgqsvzonvzrtj      others  1240000             10.0      3              3      3
```

```
143631 rows x 7 columns
```

Encoding company\_hash, job\_position

```
encoder= LabelEncoder()
```

```
X['company_hash']=encoder.fit_transform(X['company_hash'])
```

```
X['job_position']=encoder.fit_transform(X['job_position'])
```

```
X
```

|        | company_hash | job_position | ctc     | years_of_experience | class | designation | tier |
|--------|--------------|--------------|---------|---------------------|-------|-------------|------|
| 0      | 53           | 368          | 1100000 | 8.0                 | 1     | 3           | 2    |
| 1      | 1819         | 227          | 449999  | 6.0                 | 3     | 3           | 3    |
| 2      | 1433         | 100          | 2000000 | 9.0                 | 3     | 3           | 3    |
| 3      | 1123         | 100          | 700000  | 7.0                 | 3     | 3           | 3    |
| 4      | 1861         | 227          | 1400000 | 7.0                 | 1     | 3           | 1    |
| ...    | ...          | ...          | ...     | ...                 | ...   | ...         | ...  |
| 143626 | 1061         | 368          | 2250000 | 13.0                | 1     | 1           | 1    |
| 143627 | 2633         | 368          | 220000  | 16.0                | 3     | 3           | 3    |
| 143628 | 766          | 368          | 500000  | 7.0                 | 3     | 3           | 3    |
| 143629 | 2665         | 368          | 700000  | 3.0                 | 3     | 3           | 3    |
| 143630 | 152          | 368          | 1240000 | 10.0                | 3     | 3           | 3    |

143631 rows × 7 columns

Scaling all the features to start modelling

```
from sklearn.preprocessing import MinMaxScaler
ms = MinMaxScaler()
# ----- fix X #do this after label encoding
X = pd.DataFrame(ms.fit_transform(X), columns=X.columns)
X
```

|        | company_hash | job_position | ctc      | years_of_experience | class | designation | tier |
|--------|--------------|--------------|----------|---------------------|-------|-------------|------|
| 0      | 0.014838     | 0.499322     | 0.330330 | 0.444444            | 0.0   | 1.0         | 0.5  |
| 1      | 0.509239     | 0.308005     | 0.135134 | 0.333333            | 1.0   | 1.0         | 1.0  |
| 2      | 0.401176     | 0.135685     | 0.600600 | 0.500000            | 1.0   | 1.0         | 1.0  |
| 3      | 0.314390     | 0.135685     | 0.210210 | 0.388889            | 1.0   | 1.0         | 1.0  |
| 4      | 0.520997     | 0.308005     | 0.420420 | 0.388889            | 0.0   | 1.0         | 0.0  |
| ...    | ...          | ...          | ...      | ...                 | ...   | ...         | ...  |
| 143626 | 0.297032     | 0.499322     | 0.675675 | 0.722222            | 0.0   | 0.0         | 0.0  |
| 143627 | 0.737122     | 0.499322     | 0.066066 | 0.888889            | 1.0   | 1.0         | 1.0  |
| 143628 | 0.214446     | 0.499322     | 0.150150 | 0.388889            | 1.0   | 1.0         | 1.0  |
| 143629 | 0.746081     | 0.499322     | 0.210210 | 0.166667            | 1.0   | 1.0         | 1.0  |
| 143630 | 0.042553     | 0.499322     | 0.372372 | 0.555556            | 1.0   | 1.0         | 1.0  |

143631 rows × 7 columns

```
# install pylustertend
!pip install pylustertend
```

```
Collecting pylustertend
  Using cached pylustertend-1.6.2-py3-none-any.whl.metadata (3.3 kB)
Requirement already satisfied: matplotlib<4.0.0,>=3.3.3 in /usr/local/lib/python3.10/dist-packages (from pylustertend)
Requirement already satisfied: numpy<2.0.0,>=1.19.1 in /usr/local/lib/python3.10/dist-packages (from pylustertend) (1.2)
Collecting pandas<2.0.0,>=1.2.0 (from pylustertend)
  Using cached pandas-1.5.3-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (11 kB)
Collecting scikit-learn<0.25.0,>=0.24.0 (from pylustertend)
  Using cached scikit-learn-0.24.2.tar.gz (7.5 MB)
Installing build dependencies ... done
Getting requirements to build wheel ... done
error: subprocess-exited-with-error

× Preparing metadata (pyproject.toml) did not run successfully.
  exit code: 1
  See above for output.

note: This error originates from a subprocess, and is likely not a problem with pip.
Preparing metadata (pyproject.toml) ... error
error: metadata-generation-failed

× Encountered error while generating package metadata.
  See above for output.

note: This is an issue with the package mentioned above, not pip.
hint: See above for details.
```

```
from pyclustertend import hopkins
hop=hopkins(X,150)
```



```
ModuleNotFoundError                                Traceback (most recent call last)
<ipython-input-191-c1e094fea9b5> in <cell line: 1>()
----> 1 from pyclustertend import hopkins
      2 hop=hopkins(X,150)
```

**ModuleNotFoundError:** No module named 'pyclustertend'

NOTE: If your import is failing due to a missing package, you can manually install dependencies using either `!pip` or `!apt`.

To view examples of installing some common dependencies, click the "Open Examples" button below.

OPEN EXAMPLES

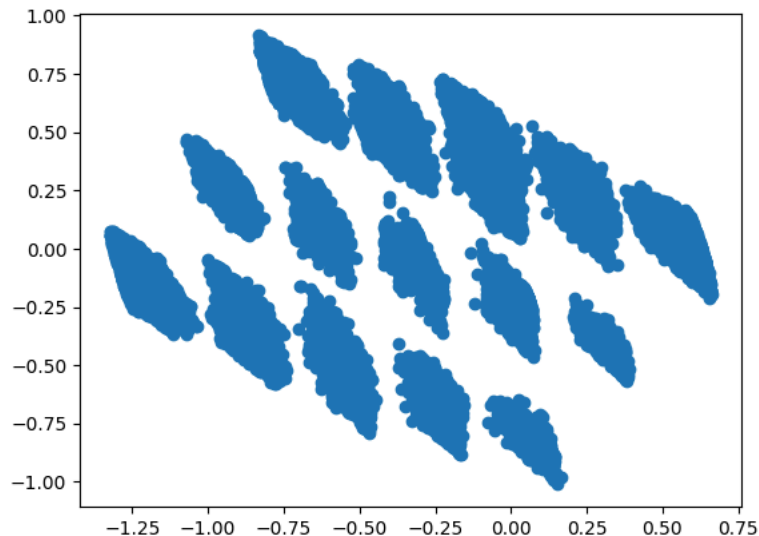
Next steps: [Explain error](#)

Not able to install hopkins in colab, so checking clustering tendency using PCA

```
# Use PCA to check the scatter plot of the data
from sklearn.decomposition import PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X)
plt.scatter(X_pca[:, 0], X_pca[:, 1])
```

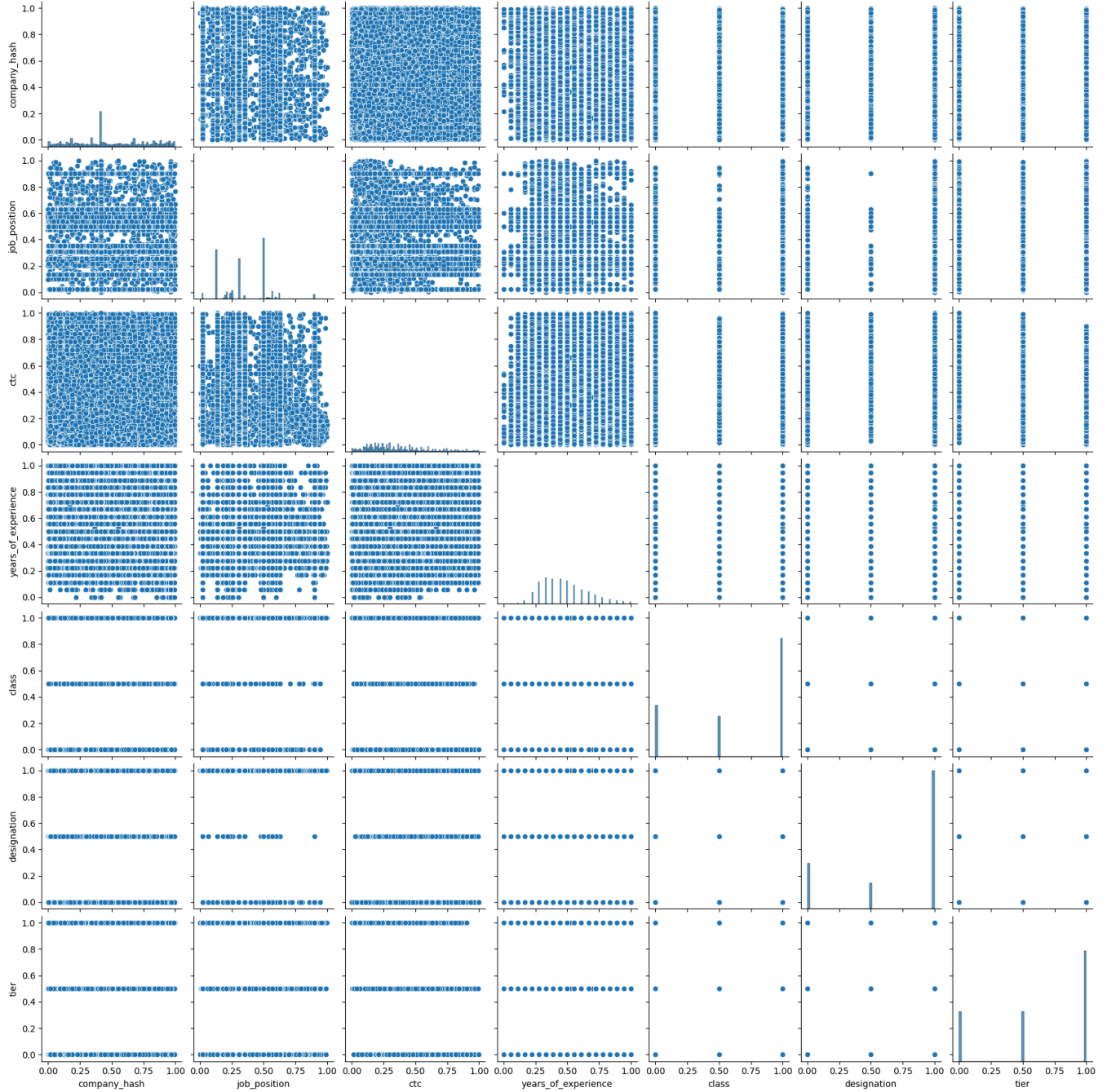


<matplotlib.collections.PathCollection at 0x7caaa9bd32e0>



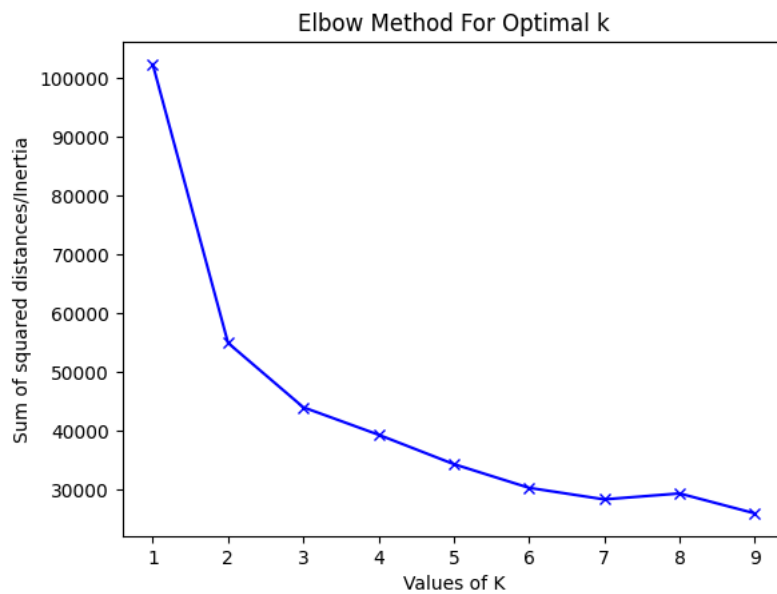
```
sns.pairplot(X)
```

 <seaborn.axisgrid.PairGrid at 0x7caae3d97e0>



Find optimal K value to perform K-Means Clustering.

```
Sum_of_squared_distances = []
K = range(1,10)
for num_clusters in K :
    kmeans = KMeans(n_clusters=num_clusters)
    kmeans.fit(X)
    Sum_of_squared_distances.append(kmeans.inertia_)
plt.plot(K,Sum_of_squared_distances,'bx-')
plt.xlabel('Values of K')
plt.ylabel('Sum of squared distances/Inertia')
plt.title('Elbow Method For Optimal k')
plt.show()
```



Obeservation: Optimal value of k is 3 clusters from the elbow method

```
kmeans = KMeans(n_clusters=3)

kmeans.fit(X)
print(kmeans.cluster_centers_)
print(kmeans.cluster_centers_.shape)
```

[[0.49843649 0.33327486 0.45935568 0.56393488 0.37183607 0.85847642  
0.31793836]  
[0.49798597 0.3420626 0.53380816 0.48543511 0.13959597 0.03988456  
0.18881935]  
[0.4975257 0.35402014 0.21931378 0.43750165 0.9780826 0.92910546  
0.95428051]]  
(3, 7)

```
X['k-m label']=kmeans.fit_predict(X)
```

X

|        | company_hash | job_position | ctc      | years_of_experience | class | designation | tier | k-m | label |
|--------|--------------|--------------|----------|---------------------|-------|-------------|------|-----|-------|
| 0      | 0.014838     | 0.499322     | 0.330330 | 0.444444            | 0.0   | 1.0         | 0.5  | 2   |       |
| 1      | 0.509239     | 0.308005     | 0.135134 | 0.333333            | 1.0   | 1.0         | 1.0  | 0   |       |
| 2      | 0.401176     | 0.135685     | 0.600600 | 0.500000            | 1.0   | 1.0         | 1.0  | 0   |       |
| 3      | 0.314390     | 0.135685     | 0.210210 | 0.388889            | 1.0   | 1.0         | 1.0  | 0   |       |
| 4      | 0.520997     | 0.308005     | 0.420420 | 0.388889            | 0.0   | 1.0         | 0.0  | 2   |       |
| ...    | ...          | ...          | ...      | ...                 | ...   | ...         | ...  | ... | ...   |
| 143626 | 0.297032     | 0.499322     | 0.675675 | 0.722222            | 0.0   | 0.0         | 0.0  | 1   |       |
| 143627 | 0.737122     | 0.499322     | 0.066066 | 0.888889            | 1.0   | 1.0         | 1.0  | 0   |       |
| 143628 | 0.214446     | 0.499322     | 0.150150 | 0.388889            | 1.0   | 1.0         | 1.0  | 0   |       |
| 143629 | 0.746081     | 0.499322     | 0.210210 | 0.166667            | 1.0   | 1.0         | 1.0  | 0   |       |
| 143630 | 0.042553     | 0.499322     | 0.372372 | 0.555556            | 1.0   | 1.0         | 1.0  | 0   |       |

143631 rows × 8 columns

Questions:

```
# What percentage users fall into largest cluster
X['k-m label'].value_counts(normalize=True)*100
```

| proportion |           |
|------------|-----------|
| k-m        | label     |
| 0          | 55.781830 |
| 2          | 22.747179 |
| 1          | 21.470992 |

dtype: float64

1)What percentage of users fall into largest cluster?

Over 56% percentage of users fall into the largest cluster.

```
Z=X.sample(frac=0.025)
Z.drop(columns=['k-m label'],inplace=True)
Z
```

|        | company_hash | job_position | ctc      | years_of_experience | class | designation | tier |
|--------|--------------|--------------|----------|---------------------|-------|-------------|------|
| 42150  | 0.061870     | 0.308005     | 0.165165 | 0.277778            | 1.0   | 1.0         | 1.0  |
| 83068  | 0.719485     | 0.499322     | 0.285285 | 0.388889            | 1.0   | 1.0         | 1.0  |
| 64324  | 0.085386     | 0.308005     | 0.264264 | 0.444444            | 1.0   | 1.0         | 1.0  |
| 25282  | 0.573908     | 0.499322     | 0.084084 | 0.444444            | 1.0   | 1.0         | 1.0  |
| 105431 | 0.416853     | 0.302578     | 0.396396 | 0.611111            | 0.0   | 0.5         | 0.5  |
| ...    | ...          | ...          | ...      | ...                 | ...   | ...         | ...  |
| 34623  | 0.179171     | 0.559023     | 0.114114 | 0.500000            | 1.0   | 1.0         | 1.0  |
| 134191 | 0.798152     | 0.629579     | 0.258258 | 0.500000            | 0.0   | 1.0         | 1.0  |
| 67817  | 0.416853     | 0.135685     | 0.900901 | 0.333333            | 0.0   | 0.0         | 0.0  |
| 136089 | 0.968925     | 0.308005     | 0.300300 | 0.722222            | 1.0   | 1.0         | 1.0  |
| 125991 | 0.955767     | 0.135685     | 0.198198 | 0.388889            | 1.0   | 1.0         | 1.0  |

3591 rows × 7 columns

Next steps:

Generate code with Z

View recommended plots

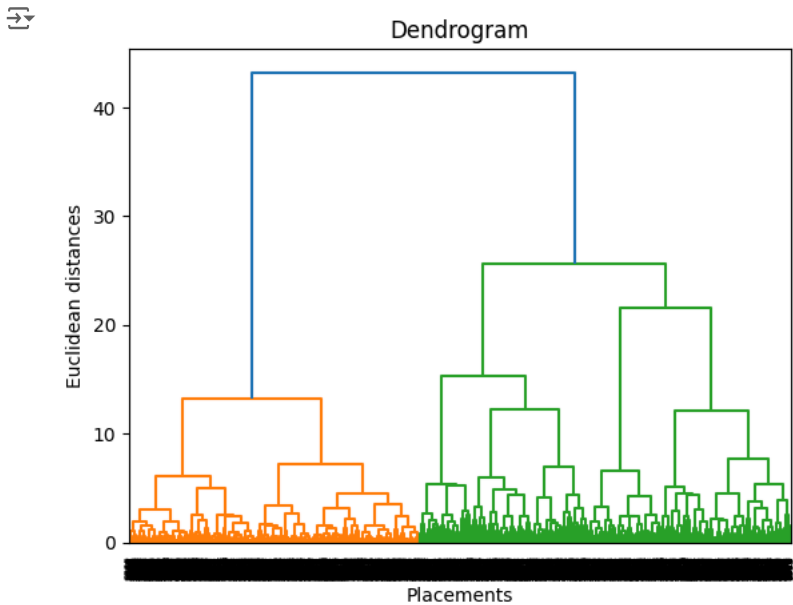
New interactive sheet

```
import sys
sys.setrecursionlimit(100000)
```

Dendrogram



```
import scipy.cluster.hierarchy as sch
dendrogram = sch.dendrogram(sch.linkage(Z, method = 'ward'))
plt.title('Dendrogram')
plt.xlabel('Placements')
plt.ylabel('Euclidean distances')
plt.show()
```



```
# Agglomerative clustering

from sklearn.cluster import AgglomerativeClustering
cluster = AgglomerativeClustering(n_clusters=3, linkage='ward')
Z['Aglo-label']=cluster.fit_predict(Z)
```

Z

|        | company_hash | job_position | ctc      | years_of_experience | class | designation | tier | Aglo-label |  |
|--------|--------------|--------------|----------|---------------------|-------|-------------|------|------------|--|
| 42150  | 0.061870     | 0.308005     | 0.165165 | 0.277778            | 1.0   | 1.0         | 1.0  | 2          |  |
| 83068  | 0.719485     | 0.499322     | 0.285285 | 0.388889            | 1.0   | 1.0         | 1.0  | 2          |  |
| 64324  | 0.085386     | 0.308005     | 0.264264 | 0.444444            | 1.0   | 1.0         | 1.0  | 2          |  |
| 25282  | 0.573908     | 0.499322     | 0.084084 | 0.444444            | 1.0   | 1.0         | 1.0  | 2          |  |
| 105431 | 0.416853     | 0.302578     | 0.396396 | 0.611111            | 0.0   | 0.5         | 0.5  | 1          |  |
| ...    | ...          | ...          | ...      | ...                 | ...   | ...         | ...  | ...        |  |
| 34623  | 0.179171     | 0.559023     | 0.114114 | 0.500000            | 1.0   | 1.0         | 1.0  | 2          |  |
| 134191 | 0.798152     | 0.629579     | 0.258258 | 0.500000            | 0.0   | 1.0         | 1.0  | 1          |  |
| 67817  | 0.416853     | 0.135685     | 0.900901 | 0.333333            | 0.0   | 0.0         | 0.0  | 0          |  |
| 136089 | 0.968925     | 0.308005     | 0.300300 | 0.722222            | 1.0   | 1.0         | 1.0  | 2          |  |
| 125991 | 0.955767     | 0.135685     | 0.198198 | 0.388889            | 1.0   | 1.0         | 1.0  | 2          |  |

3591 rows x 8 columns

Next steps:

Generate code with Z

View recommended plots

New interactive sheet

Insights and Recommendations

Insights from Manual Clustering:

- Employee Segmentation: Manually clustering based on company\_hash, job\_position, and years\_of\_experience reveals distinct segments within the learner population. Creating designations (designation and class) provides a framework for categorizing employees based on their relative positions within their companies.
- Top Performers and Companies: Identifying the top earners (Tier 1) and companies (based on CTC) highlights potential areas for Scaler to focus on. This insight allows for targeted initiatives towards specific segments.
- Job Position Analysis: Analyzing the most frequent job positions associated with Tier 1 and Class 1 learners reveals the desired roles and skill sets within the learner community. This understanding can inform curriculum development and job placement support.

### Insights from Unsupervised Clustering

- **Clustering Tendency:** Although the PCA and pairplot showed some limitations, they indicated a potential for clustering within the data. This suggests the possibility of grouping learners with similar characteristics.
- **Optimal Clusters:** The elbow method suggested 3 as the optimal number of clusters for K-Means. This provides a starting point for understanding different learner segments using unsupervised learning.
- **Cluster Distribution:** K-Means analysis revealed that over 56% of users belong to the largest cluster. Exploring the characteristics of this dominant cluster is crucial to understanding a significant portion of Scaler's learner base. **Hierarchical Structure:** While performed on a sample, hierarchical clustering offered a visual depiction of the relationships between learners based on the selected features. This perspective can provide complementary insights to K-Means. Okay, here are some recommendations for the business based on the analysis and modeling done in the code:

### Recommendations for Scaler:

- **Targeted Curriculum Development:**
  1. Focus on curriculum development for in-demand roles and skill sets identified through the analysis of top-tier learners and their preferred job positions. Specifically, job positions such as data scientist, software engineer, and data analyst emerged as popular among top earners. Tailor curriculum offerings to align with industry trends and equip learners for success in these desired roles.
  2. Address the skills gap by designing specialized courses that specifically address the needs of the industry. By focusing on industry-relevant skills and technologies, Scaler can enhance the employability of its learners and increase their chances of getting hired by top-tier companies.
- **Job Placement Support:**
  1. Leverage the insights on top-tier companies and job positions to guide learners towards their career aspirations. Partner with companies that are known to offer high compensation and career growth opportunities.
  2. Develop personalized job recommendations for learners based on their skills, experience, and cluster affiliations. By aligning learners with suitable positions, Scaler can improve their chances of finding fulfilling employment. **Strategic Partnerships:**
  3. Foster strategic partnerships with leading companies to create internship or job placement opportunities. Collaborate with industry leaders to ensure that Scaler's curriculum stays updated with the latest advancements and that graduates have access to relevant employment opportunities.
  4. Build strong relationships with companies to ensure a steady pipeline of job opportunities for graduates. By maintaining strong connections with industry partners, Scaler can provide its learners with a competitive edge in the job market.
- **Focus on Data Science Careers:**
  1. Given the prominence of data science roles among top earners, concentrate on offering comprehensive training and resources for data science-related positions. Offer specialized mentorship programs for aspiring data scientists.
  2. Promote data science career paths by highlighting success stories and providing resources for skill development. Emphasize the growing demand for data scientists and provide learners with the necessary support to transition into this field.
- **Community Building:**
  1. Facilitate a strong community among learners, alumni, and industry professionals to support networking and career development. Create online and offline opportunities for learners to connect with each other, industry experts, and potential employers.
  2. Organize industry events and workshops to help learners gain exposure and stay updated with current trends. By creating a strong sense of community, Scaler can provide learners with a supportive network that can assist them throughout their careers.

Start coding or [generate](#) with AI.

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